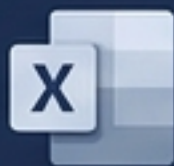


SaaS Customer Retention & Revenue Intelligence

End-to-End Data Analytics Portfolio Project

Author: Alin Kumar



Microsoft Excel



Python (Pandas)



SQL



Microsoft Power BI / DAX

Diagnosing SaaS Retention and Recurring Revenue Friction

The Business Problem

Subscription-based businesses rely on recurring revenue momentum. High churn rates, particularly in specific customer segments or acquisition channels, degrade Monthly Recurring Revenue (MRR) and increase customer acquisition costs.

Analytical Objectives

Map the complete subscription lifecycle, segment churn behavior across four product tiers, isolate high-risk cohorts, and build a scalable business intelligence model to monitor retention.

5,000

Total
Subscriptions

4,035

Active
Subscriptions

965

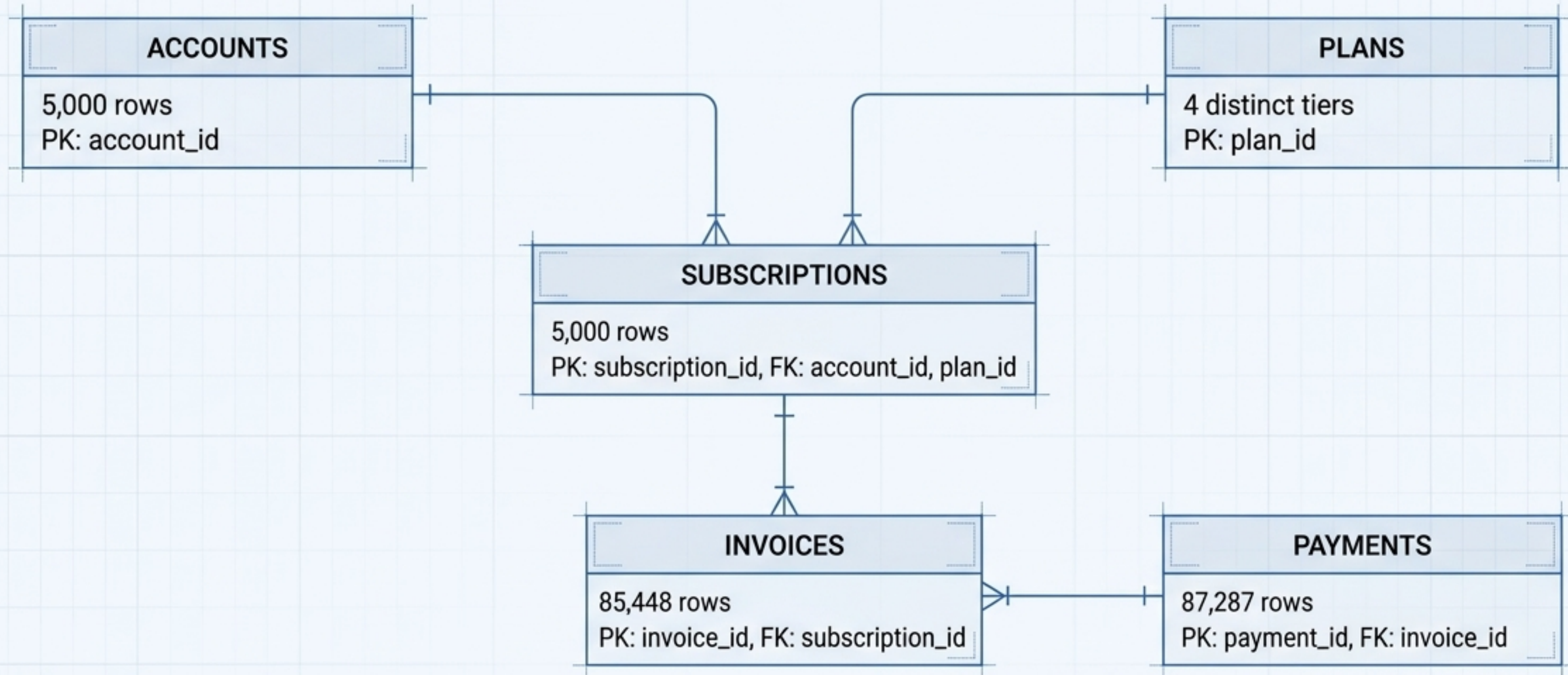
Cancelled
Subscriptions

19.3%

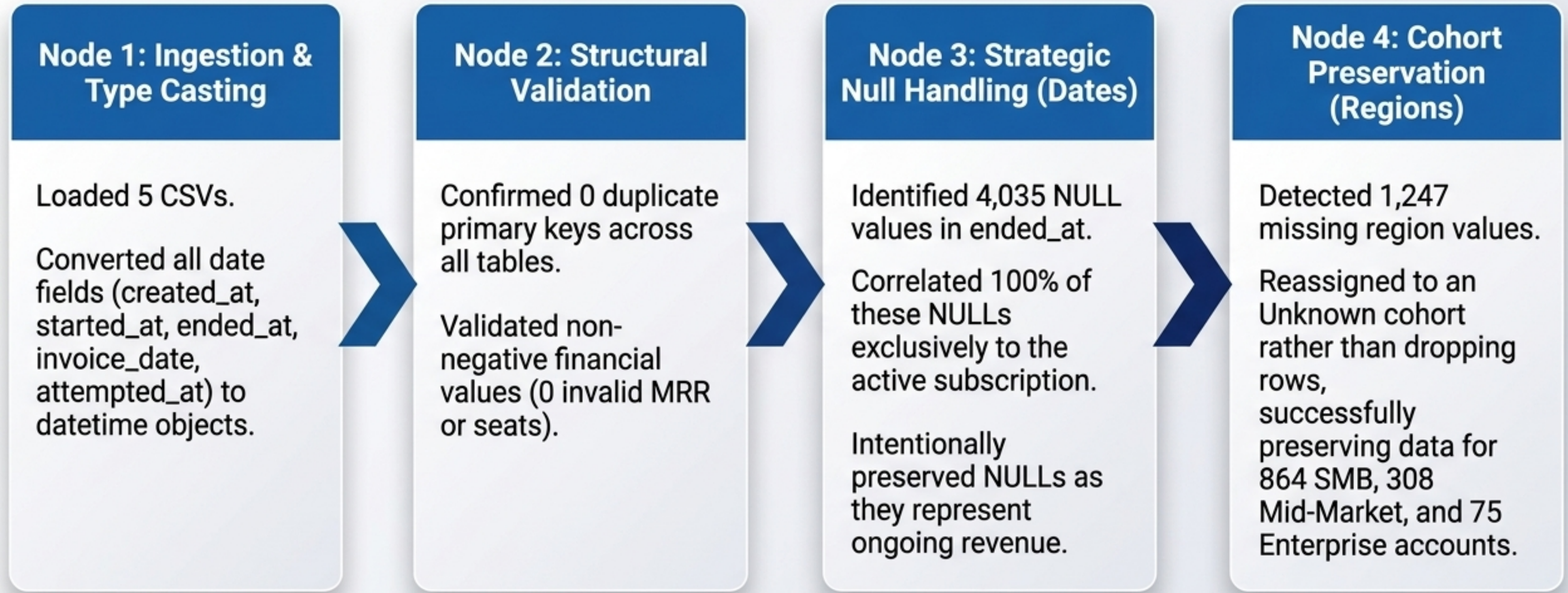
Baseline
Churn Rate

Mapping the Relational Data Architecture

Dataset Scale Summary: Analysis spans 5 interconnected datasets representing the complete billing and subscription lifecycle.



Engineered Data Quality and Validation Workflow



Deploying a Three-Phased Analytical Framework

Phase 1: Scale & Revenue Basics

Quantifying total accounts, mapping Active vs. Cancelled subscriptions, and establishing baseline revenue parameters by plan type.

Phase 2: Churn Diagnostics

Slicing the overall churn rate across critical business dimensions—product plan, customer segment, acquisition channel, and customer tenure.

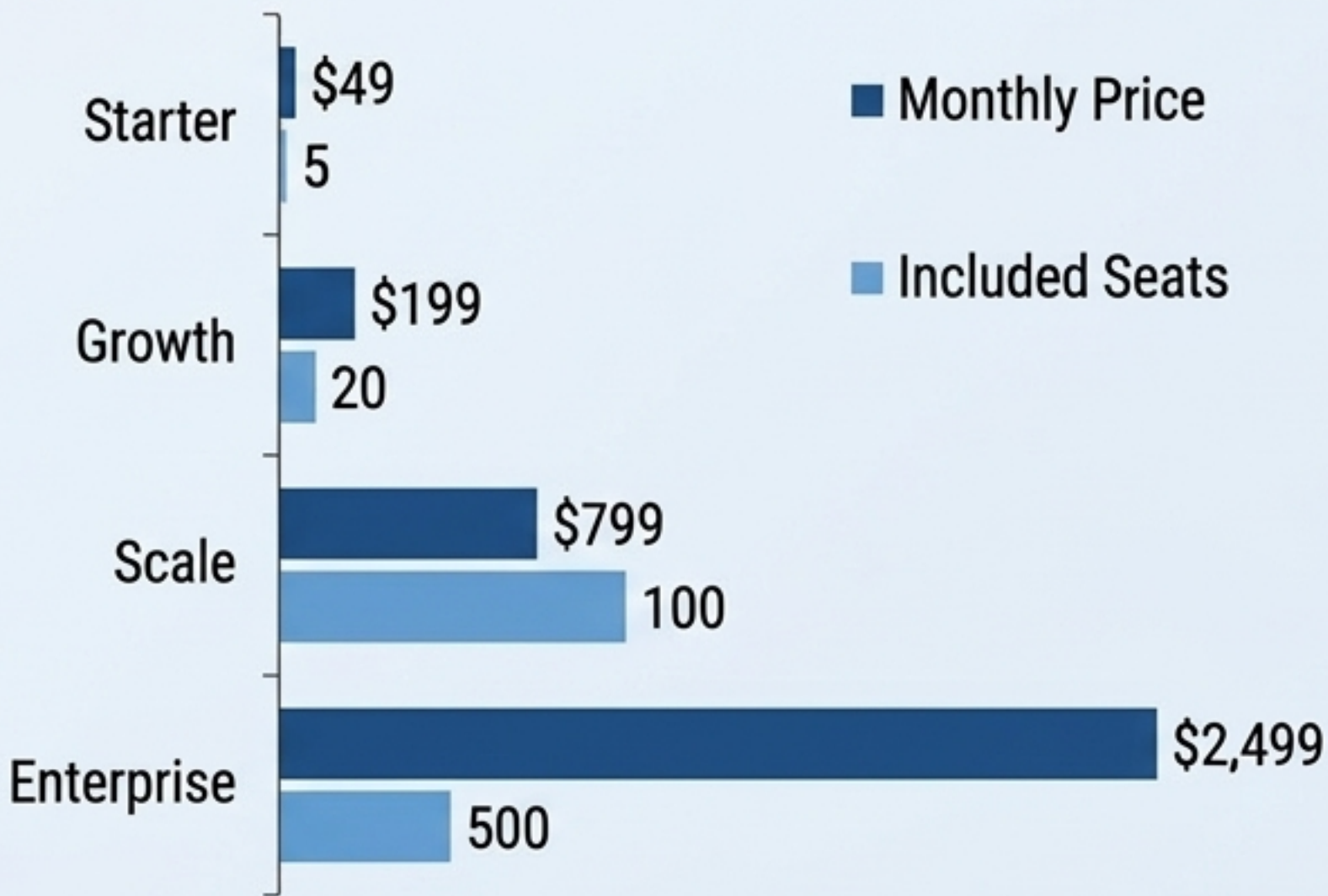
Phase 3: Root-Cause Synthesis

Executing advanced multi-dimensional queries to cross-reference risk factors and quantify the true MRR impact of specific customer profiles.

Baseline Product Economics and Plan Architecture

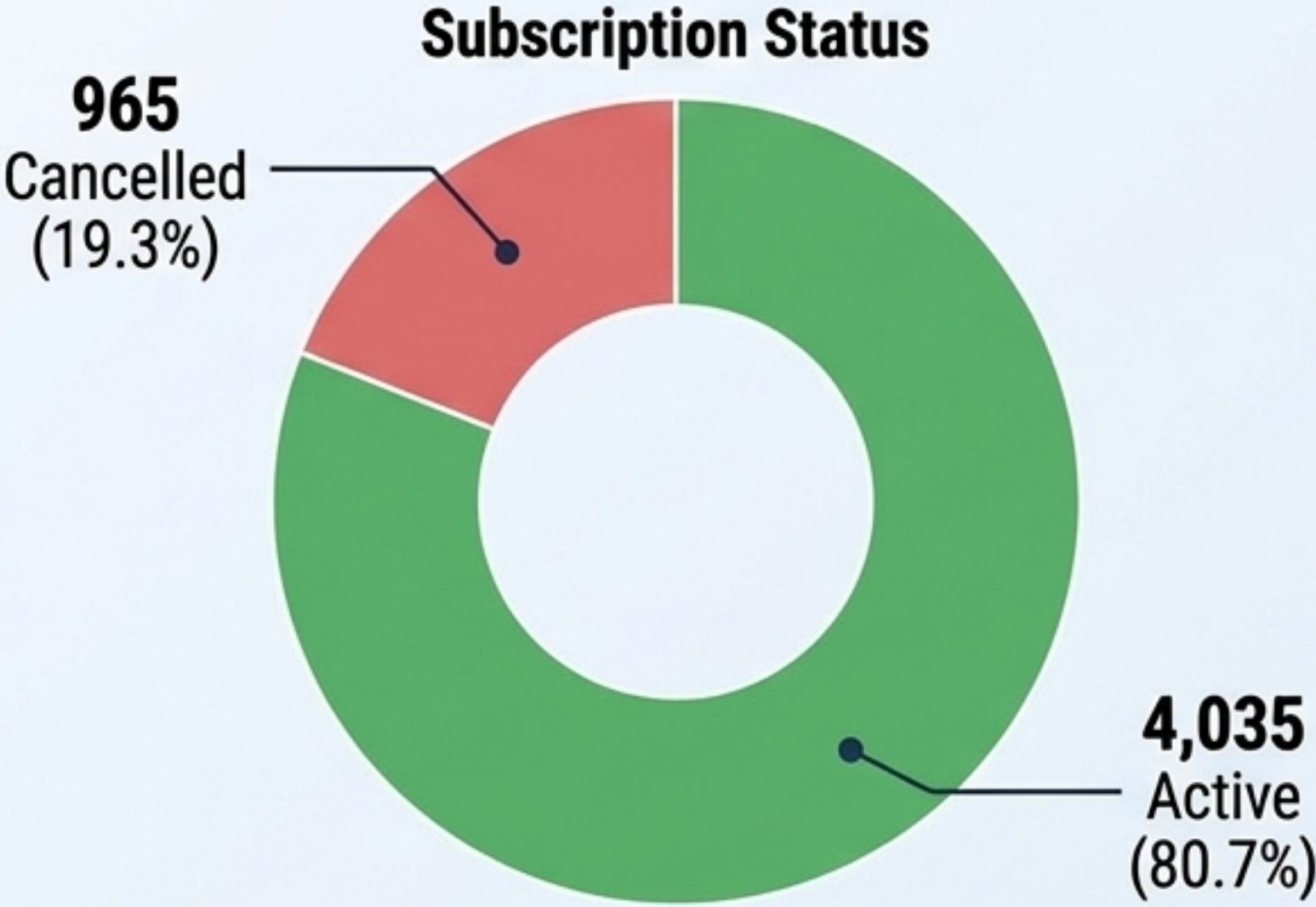
The product suite is designed to scale exponentially, moving from low-cost, low-seat entry entry points to high-value enterprise deployments.

Plan Name	Monthly Price	Included Seats
Starter Plan	\$49/month	5 Included Seats
Growth Plan	\$199/month	20 Included Seats
Scale Plan	\$799/month	100 Included Seats
Enterprise Plan	\$2,499/month	500 Included Seats



Baseline Retention and Billing System Health

Out of 5,000 total historical subscriptions, **19.3%** have **churned**. The **billing pipeline remains robust, with a highly** successful payment capture rate.



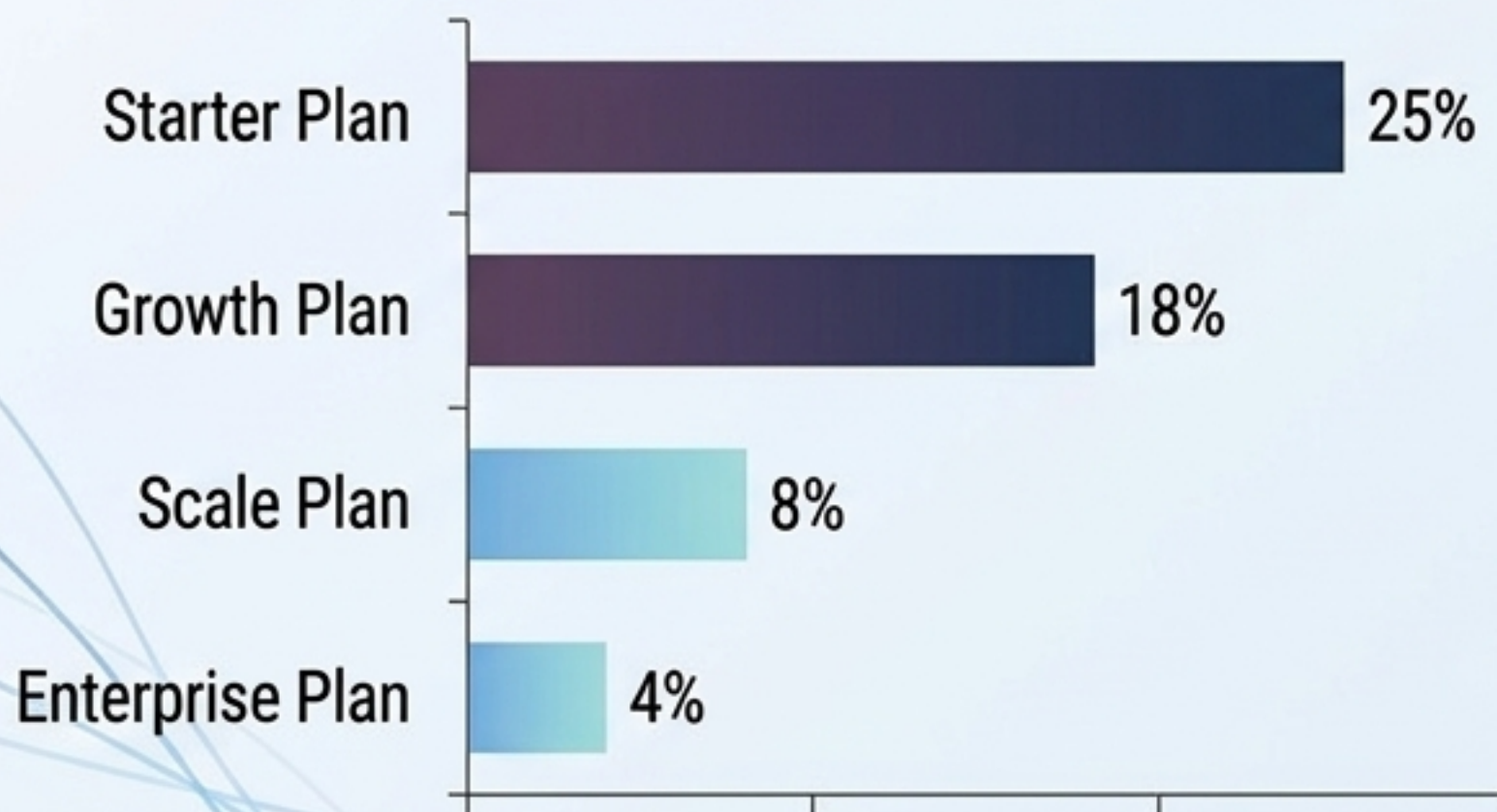
Invoice Health Pipeline		
80,101 Paid	5,311 Open	36 Void

Payment Execution	
80,101 Succeeded	7,186 Failed

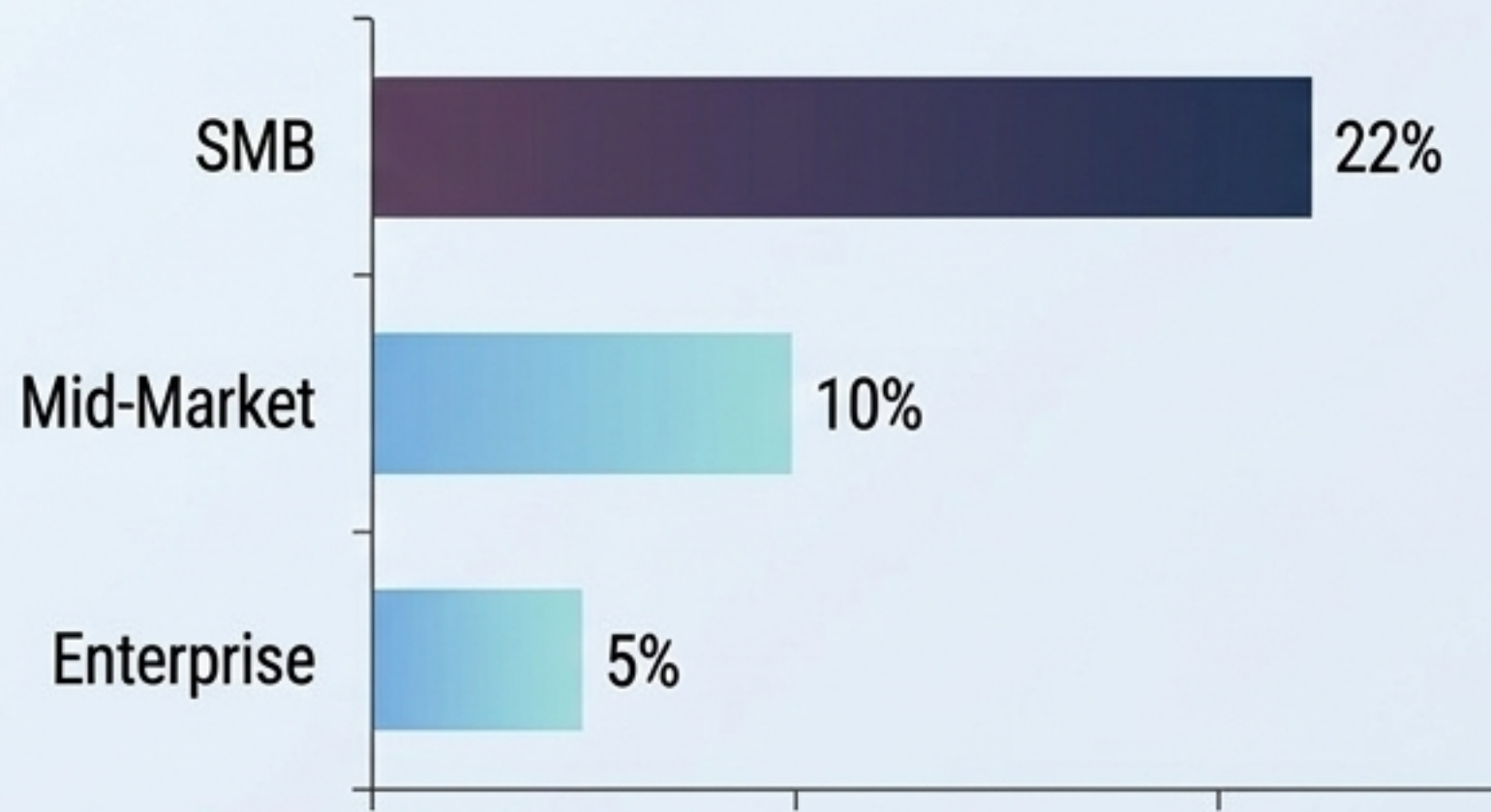
Dimensional Churn Diagnostics: Product Tier and Segment

Churn risk inversely correlates with account size and plan tier. High-tier Enterprise and Mid-Market accounts show significant stability compared to entry-level SMB users.

Churn Rate by Plan

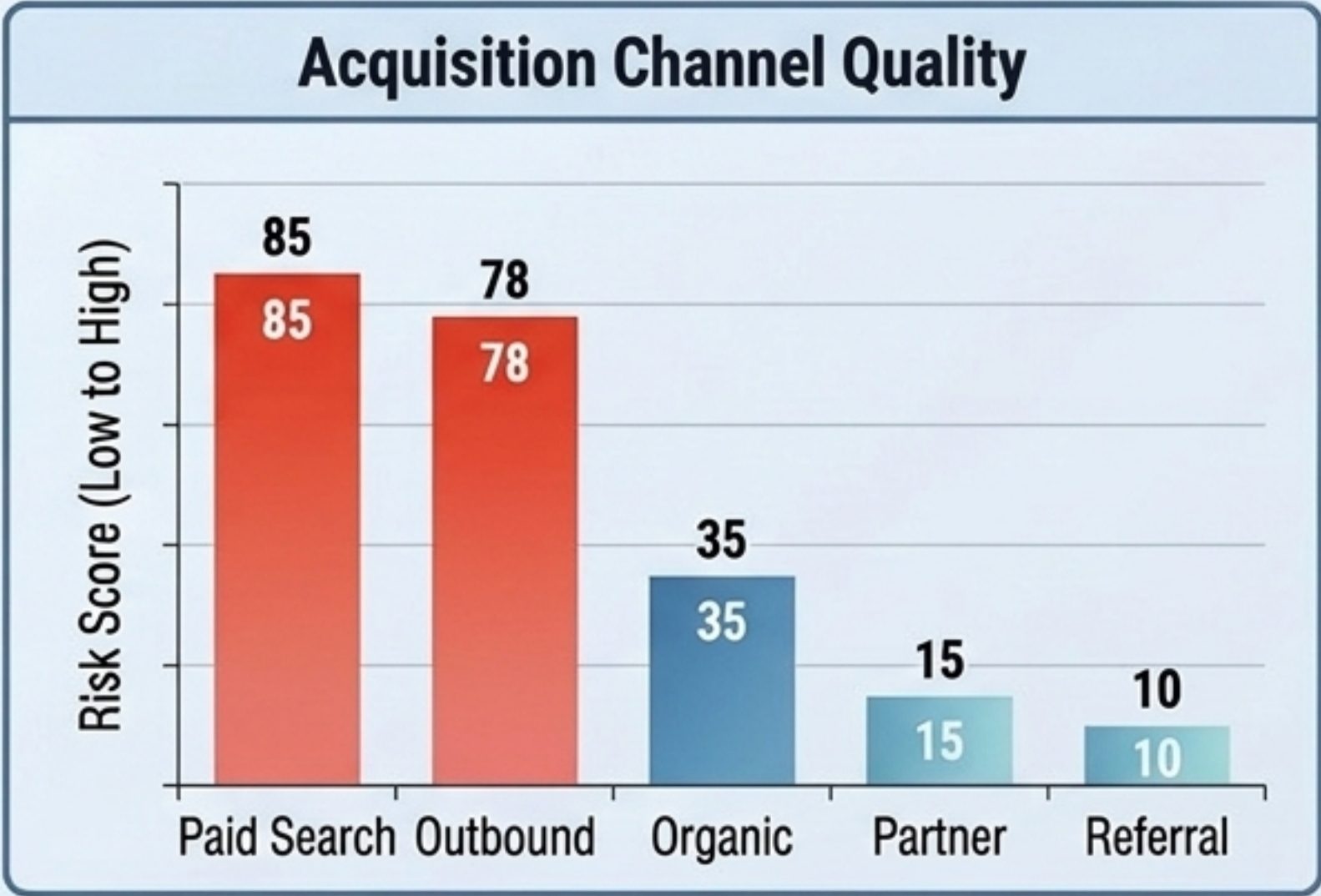
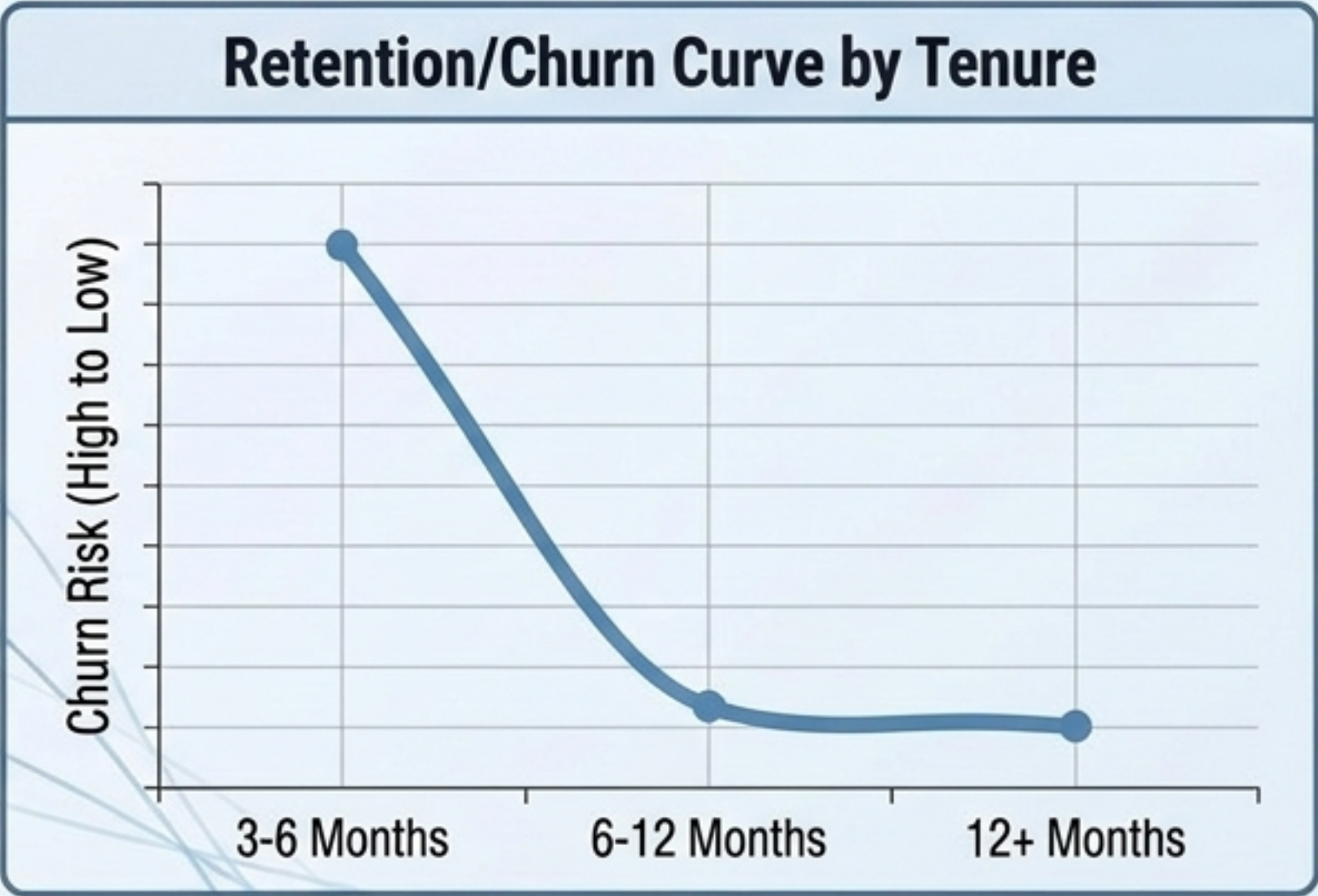


Churn Rate by Customer Segment



Dimensional Churn Diagnostics: Acquisition Channel and Tenure

Acquisition strategy dictates customer longevity. Paid channels aimed at early tenure groups demonstrate steep drop-offs, whereas organic and partner-led acquisitions foster long-term retention.



Multi-Dimensional Root-Cause Analysis

- **The Highest Risk Profiles:** 100% churn isolated entirely within the Starter Plan / SMB Segment / 3-6 Month tenure window.
- **The Lowest Risk Profiles:** Sustained retention achieved via Enterprise / Partner channel at 12+ months.

	plan_name	segment	acquisition_channel	tenure_group	total_subscriptions	cancelled_subscriptions	churn_rate	cancelled_mrr
1	Starter	SMB	paid_search	3-6 months	24	24	100% Churn	\$1,196.00
2	Starter	SMB	outbound	6-12 months	39	34	87.18% Churn	\$1,696.00
3	Scale	mid_market	outbound	12+ months	74	14	18.92% Churn	\$11,948.00
4	Enterprise	enterprise	partner	12+ months	42	6	14.29% Churn	\$15,074.00

Quantifying the True Impact of Cancelled MRR

Concept Definition: Cancelled MRR represents the loss of recurring revenue momentum associated with cancelled subscriptions. It is a SaaS momentum metric, NOT a direct accounting or balance-sheet cash loss.

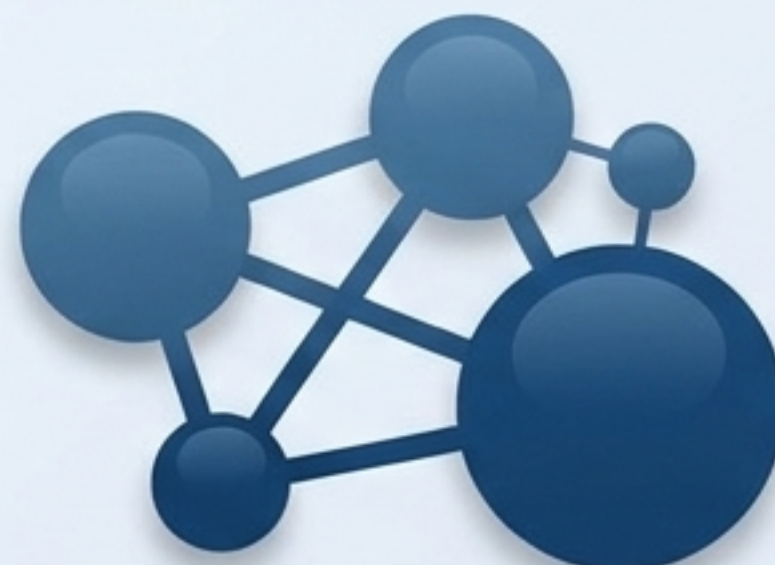
High Volume, Low MRR Impact



Starter / SMB / Paid Search (3-6m)

100% churn on 24 subscriptions
resulting in only **\$1,196 Cancelled MRR**

Low Volume, Massive MRR Impact



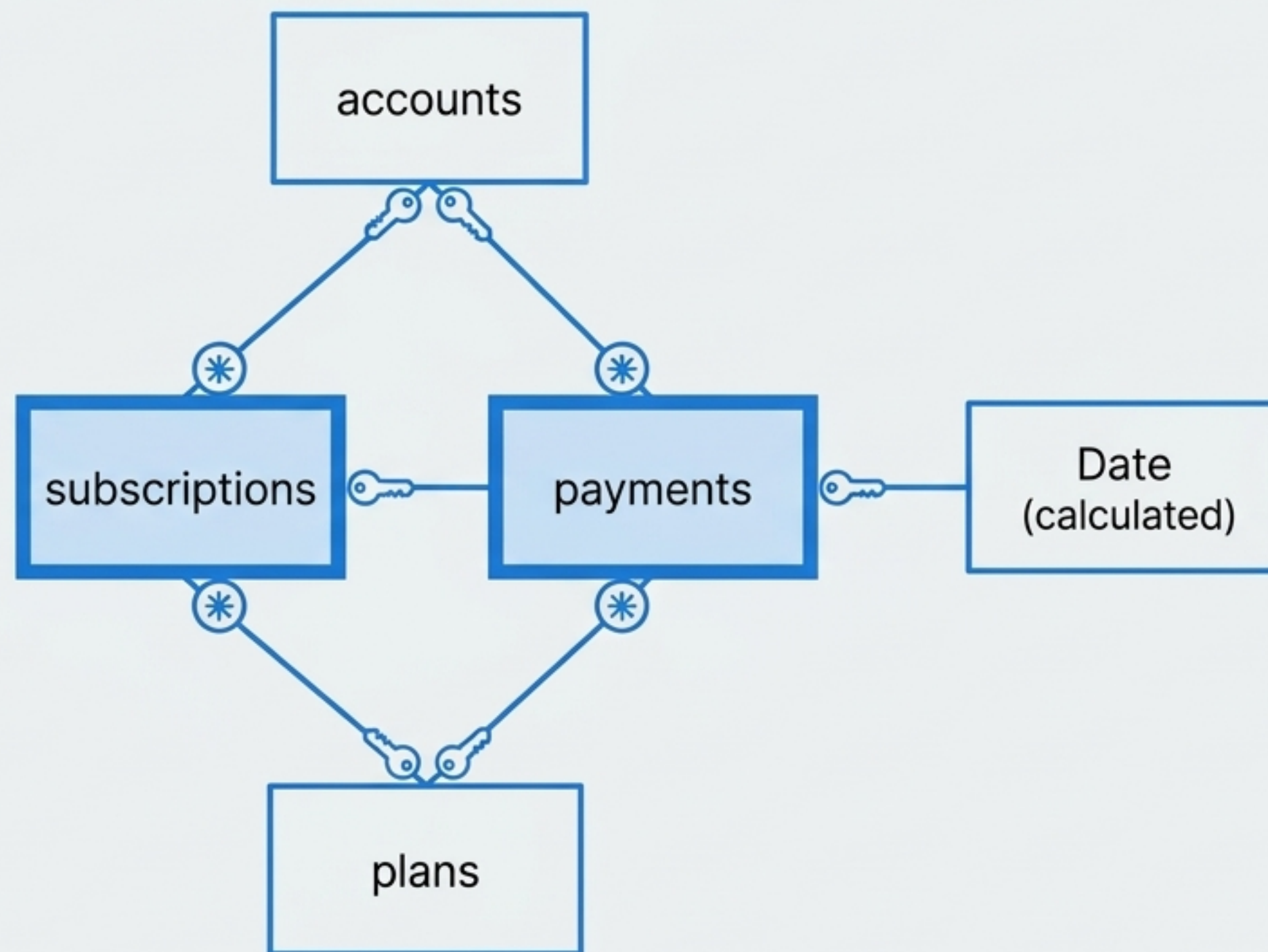
Enterprise / Partner (12+m)

14.29% churn on 6 cancelled subscriptions
resulting in a massive **\$15,074 Cancelled MRR**

Operationalizing Insights: Power BI Data Model

Architecture: Translated the cleaned Python outputs into a scalable Star Schema within Power BI.

DAX Engineering: Developed dynamic measures to calculate Active MRR, Cancelled MRR, and Churn Rates across any filtered dimension (Tenure, Segment, Channel), enabling real-time stakeholder exploration.

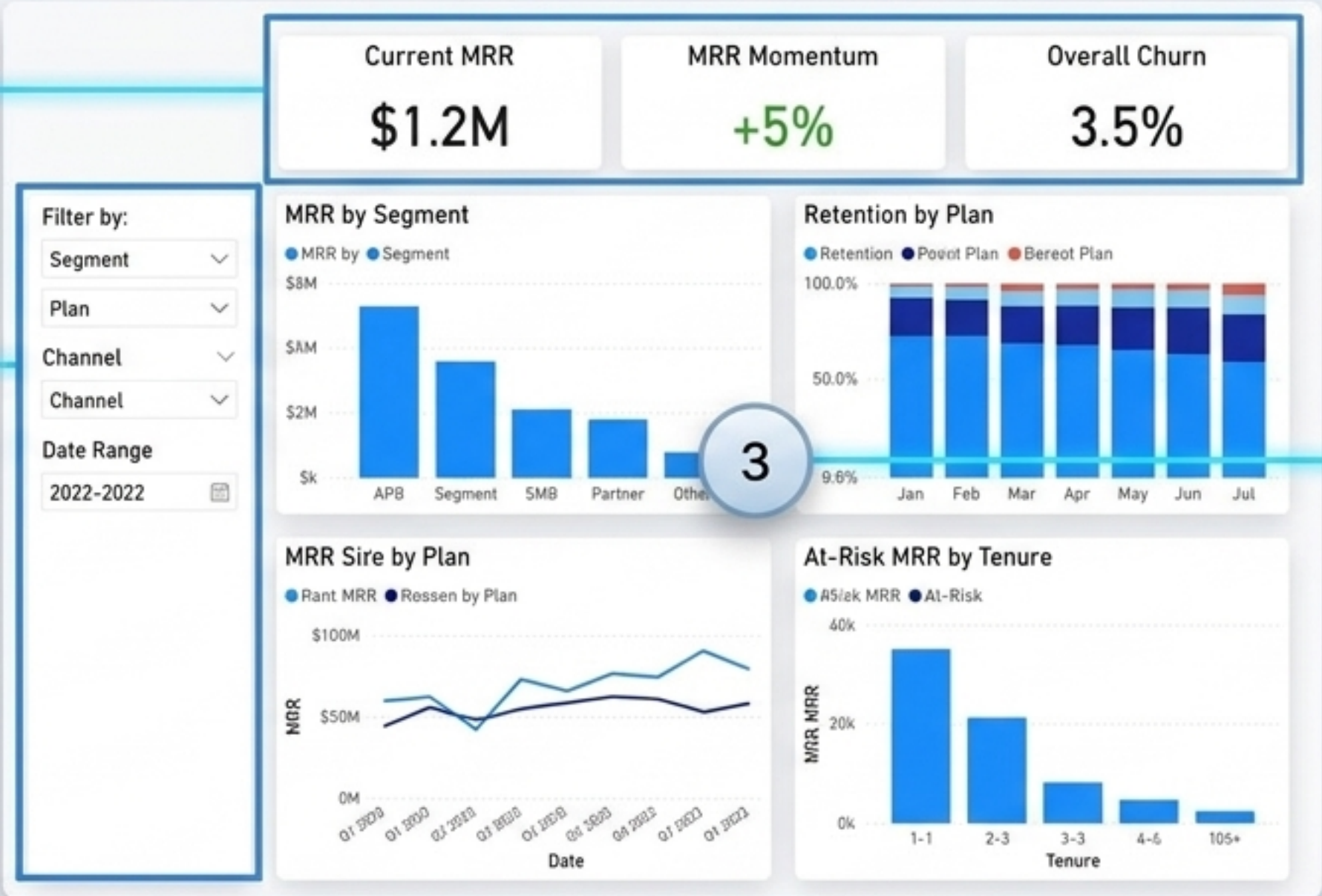


Executive SaaS Revenue & Retention Dashboard

1

2

Annotation 2:
Multi-dimensional
slicers allowing
stakeholders to drill
down into Segment,
Plan, and Channel
performance.



Annotation 1: Dynamic
KPI header tracking
vital MRR momentum
and overall churn.

Annotation 3:
Automated root-cause
visualizations
identifying at-risk
MRR by customer
tenure.

Synthesizing Evidence into Business Reality

The Data (What)

Observation: SMBs acquired via paid search on Starter plans consistently churn within their first 6 months (100% cohort churn). Conversely, Enterprise accounts via partners maintain 85%+ retention after a year.

The Insight (Why)

Interpretation: There is likely a mismatch between top-of-funnel ad messaging and the actual product value experienced by early-stage, low-tier users. Meanwhile, partner integrations create high switching costs for Enterprise clients.

The Limitation

Context: This is observational billing data, not behavioral product telemetry. It proves where churn happens, but cannot definitively prove causation without user-activity logs. Furthermore, 1,247 missing regional data points limits precise geographic targeting.

Strategic Recommendations & Project Conclusion

1. Reallocate Acquisition Spend

Divest from Paid Search targeting SMBs for the Starter plan; shift marketing budget toward nurturing Partner channels for Scale/Enterprise tiers.



2. Proactive Intervention

Implement mandatory Customer Success check-ins for all outbound/organic SMB clients at the critical 3-month tenure mark.



Portfolio Value Demonstrated

Author: Alin Kumar

- **Data Engineering:** Robust ETL and validation via Python/Pandas.
- **Advanced Analytics:** Multi-dimensional root-cause isolation using SQL.
- **Business Acumen:** Precise application of SaaS financial metrics (MRR vs. GAAP).
- **Data Visualization:** End-to-end dashboard deployment in Power BI using DAX.